The Method of Least Squares

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Challenge: Characterize Temperature Gradient

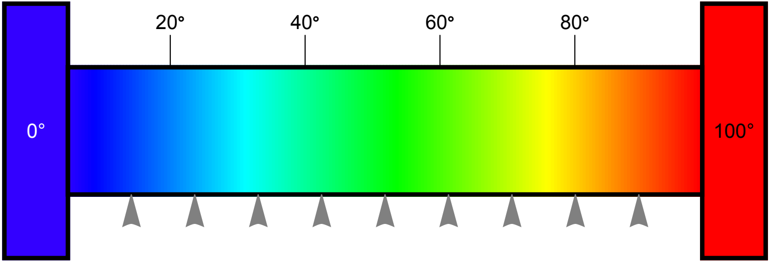


Figure: Measuring the temperature of a bar connecting two constant-temperature heat baths.

Stationary Solution of the Heat Equation

Heat Equation with Dirichlet boundary conditions

$$
\begin{aligned}
u_t &= \alpha u_{xx}, \qquad 0 \leq x \leq 10 \\
u(0) &= 0 \\
u(10) &= 100
\end{aligned} \tag{1}
$$

Stationary Solution of the Heat Equation

Equilibrium state with no temporal variation

$$u_t \Longrightarrow 0 \tag{2}$$

Reduces (1) to Laplace's equation

$$\begin{aligned} \alpha u_{xx} &= 0, \qquad 0 \leq x \leq 10 \\ u(0) &= 0 \\ u(10) &= 100 \end{aligned} \tag{3}$$

Solution of Laplace's equation

First integral:

$$u_x = \int u_{xx} dx = \int 0 dx = a_1$$

Second integral:

$$u_x = \int u_x dx = \int a_1 dx = a_0 + a_1 x$$

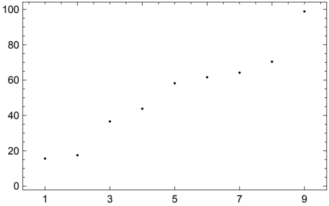
Model

Linear gradient function for the temperature distribution y as a function of position x

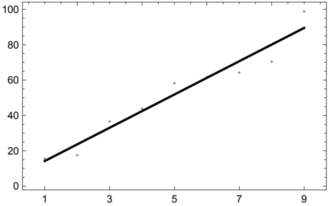
$$y(x) = a_0 + a_1 x \tag{4}$$

Best Fit to a Linear Function

input $\Rightarrow$ output



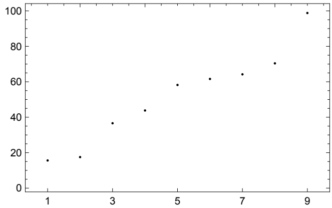
$\Rightarrow$



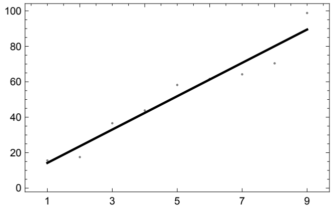
measurements $\Rightarrow$ solution

Best Fit to a Linear Function

input $\Rightarrow$ output



$\Rightarrow$

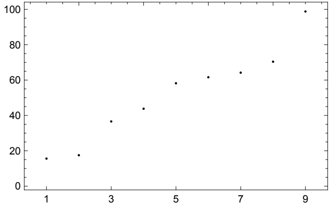


measurements $\Rightarrow$ solution

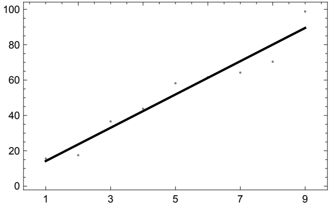
$\{$position, temperature$\}$ $\Rightarrow$ best intercept and slope

Best Fit to a Linear Function

input $\Rightarrow$ output



$\Rightarrow$



measurements $\Rightarrow$ solution

{position, temperature} $\Rightarrow$ best intercept and slope

$$\{x_k, y_k\}_{k=1}^{m} \Rightarrow \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} \pm \begin{bmatrix} \sigma_0 \\ \sigma_1 \end{bmatrix}$$

Least Squares: Ingredients

1. Model
2. Data
3. $2-$norm

Least Squares: Ingredients

1. Model: linear function $y(x) = a_0 + a_1 x$.
2. Data: position, temperature measurements, $\{x_k, y_k\}_{k=1}^{m}$
3. $2-$norm: $\|x+y\|_2^2 = x^2 + y^2$

At last...

We are ready to express the Least Squares problem.

Best Fit Using Linear Model

Model:

$$y(x) = a_0 + a_1 x \tag{5}$$

Two free parameters:

a_0: intercept

a_1: slope

Linear System

The result of m measurements:

$$
\begin{array}{ccccc}
a_0 & + & a_1 x_1 & - & y_1 \\
 & & \vdots & & \vdots \\
a_0 & + & a_1 x_m & - & y_m
\end{array}
\tag{6}
$$

Interlude: Vector Processing

The re-emergence of vector instructions started with the MMX instruction set (1997) which was added to the x86 architecture and the Intel Pentium processors

Best Fit Using Linear Model

We want to minimize the difference between model, $y(x)$, and measurements, y.

$$y(x) = a_0 + a_1 x \tag{5}$$

This difference is expressed by the residual error vector

$$r(a_0, a_1) = a_0 \mathbf{1} + a_1 x - y \tag{7}$$

Best Fit Using Linear Model

We want to minimize the magnitude of the residual error vector $r(a_0, a_1)$.

Best Fit Using Linear Model

We want to minimize $\|r(a_0, a_1)\|_2^2$.

Notation: Vectors

Three essential $m-$vectors:

$$\mathbf{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} \tag{8}$$

$$mesh \qquad\qquad data$$

Notation: Inner Product

The inner product notation $\langle x, y \rangle$ is powerful:

1. topology independent (continuous or discrete)
2. field independent ($\mathbb{R}$ or $\mathbb{C}$)

Notation: Inner Product

For $\mathbb{R}^m$, and a discrete topology l^2:

$$\begin{aligned}
\langle x, y \rangle &= x \cdot y \\
&= x^T y = \begin{bmatrix} x_1 & \dots & x_m \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} \\
&= x_1 y_1 + \cdots + x_m y_m = \sum_{k=1}^{m} x_k y_k
\end{aligned} \tag{9}$$

The result is a scalar.

Minimize Total Squared Error

The method of least squares will find the minimum value of the total squared error:

$$t^2(a_0, a_1) = \underset{a_0, a_1 \in \mathbb{R}}{\arg\min} \langle r(a_0, a_1), r(a_0, a_1) \rangle \tag{10}$$

Minimize Total Squared Error

The values of a which minimize the total squared error represent the least squares solution:

$$a_{LS} = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}_{LS} \tag{11}$$

Best Fit Using Linear Model

$$y(x) = a_0 + a_1 x \tag{5}$$

Minimize difference between prediction and measurement,

$$r(a_0, a_1) = a_0 \mathbf{1} + a_1 x - y \tag{7}$$

Minimize the total squared error:

$$t^2(a_0, a_1) = \langle r(a_0, a_1), r(a_0, a_1) \rangle \tag{10}$$

Finding the Least Squares Solution

There are many ways to minimize t^2.
We choose the fundamental tools of the calculus.

Minimization Via Calculus

How to minimize $f(u,v)$?

Find the extremal values by the simultaneous solution

$$\begin{aligned} \frac{\partial f}{\partial u} &= 0, \\ \frac{\partial f}{\partial v} &= 0. \end{aligned} \tag{12}$$

Solve by Minimizing t^2

$$\begin{aligned} t^2(a_0, a_1) &= \langle r(a_0, a_1), r(a_0, a_1)\rangle \\ &= {a_0}^2\langle \mathbf{1}, \mathbf{1}\rangle + {a_1}^2\langle x, x\rangle + \langle y, y\rangle \\ &\quad + 2a_0a_1\langle \mathbf{1}, x\rangle - 2a_0\langle \mathbf{1}, y\rangle - 2a_1\langle x, y\rangle \end{aligned} \tag{13}$$

Solve by Minimizing t^2

Find the minimum by solving the simultaneous equations

$$\begin{aligned} \frac{\partial t^2}{\partial a_0} &= 0, \\ \frac{\partial t^2}{\partial a_1} &= 0. \end{aligned} \tag{14}$$

Intercept Parameter a_0

First, the intercept parameter a_0. Three terms do not depend up a_0

$$
\begin{aligned}
\frac{\partial}{\partial a_0}\left({a_1}^2\langle x, x\rangle\right) &= 0 \\
\frac{\partial}{\partial a_0}\left(\langle y, y\rangle\right) &= 0 \\
\frac{\partial}{\partial a_0}\left(-2a_1\langle x, y\rangle\right) &= 0
\end{aligned}
\tag{15}
$$

Intercept Parameter a_0

Three terms have non-trivial derivatives:

$$\begin{aligned} \frac{\partial}{\partial a_0}\left({a_0}^2\langle \mathbf{1}, \mathbf{1}\rangle\right) &= 2a_0\langle \mathbf{1}, \mathbf{1}\rangle, \\ \frac{\partial}{\partial a_0}\left(2a_0 a_1\langle \mathbf{1}, x\rangle\right) &= 2a_1\langle \mathbf{1}, x\rangle, \\ \frac{\partial}{\partial a_0}\left(-2a_0\langle x, y\rangle\right) &= -2\langle x, y\rangle. \end{aligned} \tag{16}$$

Intercept Parameter a_0

Putting it all together:

$$\frac{\partial t^2}{\partial a_0} = 0 \quad \Rightarrow \quad a_0 \langle \mathbf{1}, \mathbf{1} \rangle + a_1 \langle \mathbf{1}, x \rangle = \langle \mathbf{1}, y \rangle \tag{17}$$

Slope Parameter a_1

Next, the intercept parameter a_1. Again, three terms do not depend up a_1

$$\begin{aligned} \frac{\partial}{\partial a_1}\left({a_0}^2\langle\mathbf{1},\mathbf{1}\rangle\right) &= 0 \\ \frac{\partial}{\partial a_1}\left(\langle y,y\rangle\right) &= 0 \\ \frac{\partial}{\partial a_1}\left(-2a_0\langle\mathbf{1},y\rangle\right) &= 0 \end{aligned} \tag{18}$$

Slope Parameter a_1

Again, three terms have non-trivial derivatives:

$$\begin{aligned} \frac{\partial}{\partial a_1}\left({a_1}^2\langle x, x\rangle\right) &= 2a_1\langle x, x\rangle, \\ \frac{\partial}{\partial a_1}\left(2a_0a_1\langle \mathbf{1}, x\rangle\right) &= 2a_0\langle \mathbf{1}, x\rangle, \\ \frac{\partial}{\partial a_1}\left(-2a_1\langle x, y\rangle\right) &= -2\langle x, y\rangle. \end{aligned} \tag{19}$$

Slope Parameter a_1

$$\frac{\partial t^2}{\partial a_0} = 0 \quad \Rightarrow \quad a_0 \langle \mathbf{1}, \mathbf{1} \rangle + a_1 \langle x, x \rangle = \langle x, y \rangle \tag{20}$$

Simultaneous Equations

Solve equations (20) and (17) simultaneously

$$\begin{aligned} a_0\langle \mathbf{1}, \mathbf{1}\rangle &+ a_1\langle \mathbf{1}, x\rangle &= \langle \mathbf{1}, y\rangle, \\ a_0\langle \mathbf{1}, x\rangle &+ a_1\langle x, x\rangle &= \langle x, y\rangle. \end{aligned} \tag{21}$$

Simultaneous Equations: Finding a_0

Solving equations (21). Start with a_0:

1. Multiply top equation by $\langle x, x \rangle$
2. Multiply bottom equation by $\langle \mathbf{1}, x \rangle$
3. Subtract bottom from top
4. Solve for a_0

Simultaneous Equations: Finding a_0

Solving equations (21). Start with a_0:

$$\begin{array}{rlcl} & a_0\langle \mathbf{1},\mathbf{1}\rangle\langle x,x\rangle & + & \cancel{a_1\langle \mathbf{1},x\rangle\langle x,x\rangle} & = & \langle \mathbf{1},y\rangle\langle x,x\rangle \\ - & a_0\langle \mathbf{1},x\rangle\langle \mathbf{1},x\rangle & + & \cancel{a_1\langle x,x\rangle\langle \mathbf{1},x\rangle} & = & \langle x,y\rangle\langle \mathbf{1},x\rangle \\ \hline \end{array}$$

$$\Downarrow$$

$$a_0\left(\langle \mathbf{1},\mathbf{1}\rangle\langle x,x\rangle - \langle \mathbf{1},x\rangle^2\right) = \langle \mathbf{1},y\rangle\langle x,x\rangle - \langle \mathbf{1},x\rangle\langle x,y\rangle$$

Simultaneous Equations: Define Determinant

$$\Delta = \langle \mathbf{1}, \mathbf{1} \rangle \langle x, x \rangle - \langle \mathbf{1}, x \rangle^2 \tag{22}$$

Simultaneous Equations: Solution for a_0

$$a_0 = \Delta^{-1} \left(\langle \mathbf{1}, y \rangle \langle x, x \rangle - \langle \mathbf{1}, x \rangle \langle x, y \rangle \right) \tag{23}$$

Simultaneous Equations: Finding a_1

Solving equations (21), now for a_1:

1. Multiply top equation by $\langle \mathbf{1}, x \rangle$
2. Multiply bottom equation by $\langle x, x \rangle$
3. Subtract bottom from top
4. Solve for a_1

Simultaneous Equations: Finding a_1

Solving equations (21), now for a_1:

$$
\begin{array}{rcrcl}
 & \cancel{a_0\langle \mathbf{1},\mathbf{1}\rangle\langle 1,x\rangle} & + & a_1\langle \mathbf{1},x\rangle\langle 1,x\rangle & = & \langle \mathbf{1},y\rangle\langle 1,x\rangle \\
- & \cancel{a_0\langle \mathbf{1},x\rangle\langle 1,1\rangle} & + & a_1\langle x,x\rangle\langle 1,1\rangle & = & \langle x,y\rangle\langle 1,1\rangle \\
\hline
\end{array}
$$

$$\Downarrow$$

$$a_1\left(\langle \mathbf{1},x\rangle^2 - \langle \mathbf{1},\mathbf{1}\rangle\langle x,x\rangle\right) = \langle \mathbf{1},x\rangle\langle \mathbf{1},y\rangle - \langle \mathbf{1},\mathbf{1}\rangle\langle x,y\rangle$$

Simultaneous Equations: Solution for a_1

$$a_1 = \Delta^{-1} \left(\langle \mathbf{1}, \mathbf{1} \rangle \langle x, y \rangle - \langle \mathbf{1}, x \rangle \langle \mathbf{1}, y \rangle \right) \tag{24}$$

Simultaneous Equations: Least Squares Solution

Using only calculus and algebra,
the least squares solution is

$$a_{LS} = \left[\begin{array}{c} a_0 \\ a_1 \end{array} \right]_{LS} = \Delta^{-1} \left[\begin{array}{c} \langle x, x \rangle \langle \mathbf{1}, y \rangle - \langle \mathbf{1}, x \rangle \langle x, y \rangle \\ \langle \mathbf{1}, \mathbf{1} \rangle \langle x, y \rangle - \langle \mathbf{1}, x \rangle \langle \mathbf{1}, y \rangle \end{array} \right] \tag{25}$$

Elegance of Least Squares

The method of least squares provides an elegant quantification of the error in the amplitudes.

Elegance of Least Squares

These errors define the number of significant digits.

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} \pm \begin{bmatrix} \sigma_0 \\ \sigma_1 \end{bmatrix} \tag{26}$$

Elegance of Least Squares

How does a perturbation in the data affect the solution?

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} \pm \begin{bmatrix} \sigma_0 \\ \sigma_1 \end{bmatrix} \tag{26}$$

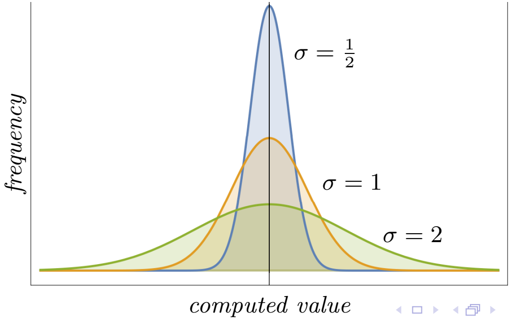
How Precise is Our Measurement?

How Precise is Our Measurement? Because of the Central Limit Theorem, we expect error to be distributed normally.

Examples: Normal distribution

Normal distribution

$$f(t;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2\right)$$



Examples: Normal distribution

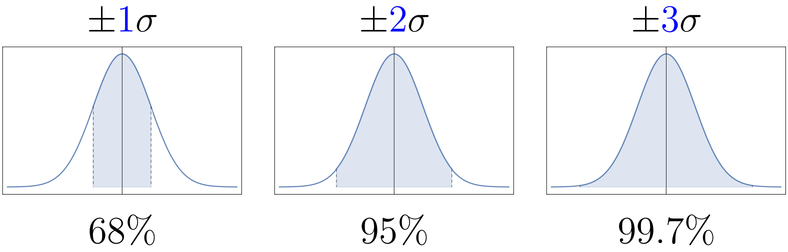
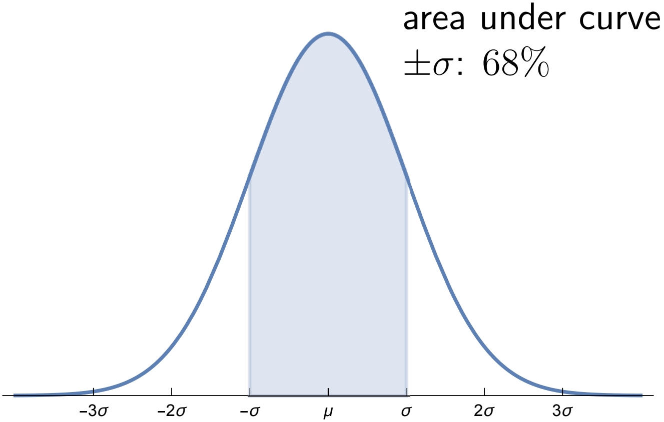
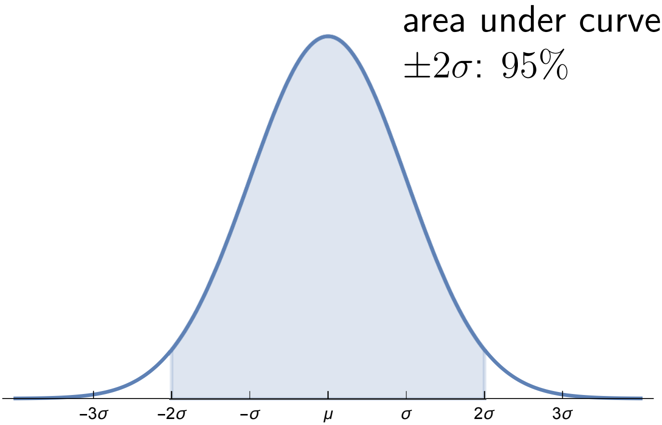


Table: $68 - 95 - 99.7$ rule

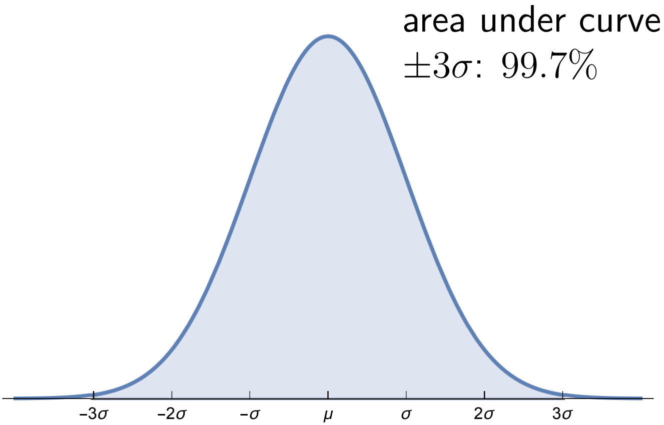
Normal Distribution: $\pm\sigma$



Normal Distribution: $\pm 2\sigma$



Normal Distribution: $\pm 3\sigma$



Least Squares Result

interval	integral		area
1σ	$\int_{-\sigma}^{\sigma} f(x;0,\sigma)\,dx$	$=$	0.682689492
2σ	$\int_{-2\sigma}^{2\sigma} f(x;0,\sigma)\,dx$	$=$	0.954499736
3σ	$\int_{-3\sigma}^{3\sigma} f(x;0,\sigma)\,dx$	$=$	0.997300204
4σ	$\int_{-4\sigma}^{4\sigma} f(x;0,\sigma)\,dx$	$=$	0.999936658
5σ	$\int_{-5\sigma}^{5\sigma} f(x;0,\sigma)\,dx$	$=$	0.999999427

Least Squares Result

Final solution

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} \pm \begin{bmatrix} \sigma_0 \\ \sigma_1 \end{bmatrix} = \begin{bmatrix} 4.8 \\ 9.41 \end{bmatrix} \pm \begin{bmatrix} 4.9 \\ 0.87 \end{bmatrix} \tag{26}$$

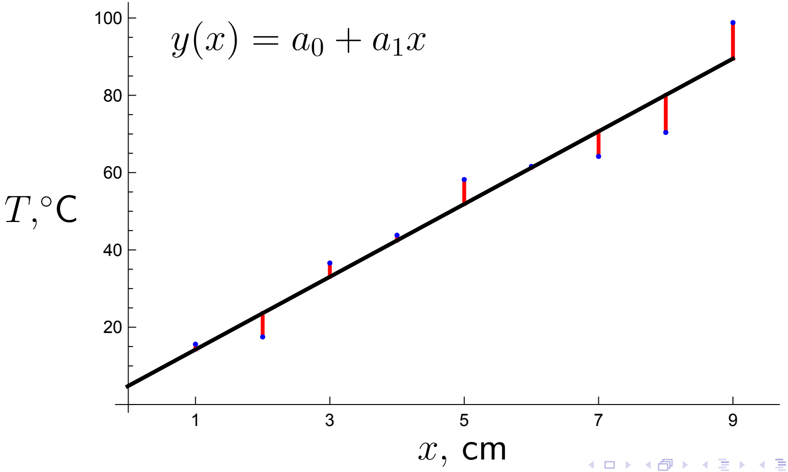
Least Squares Result

Final solution

$$\begin{array}{llcll} \text{intercept} & a_0 & = & 4.8 & \pm\ 4.9^\circ\mathrm{C} \\ \text{slope} & a_1 & = & 9.41 & \pm\ 0.87^\circ\mathrm{C}/\text{cm} \end{array} \tag{27}$$

Final digit is a guard digit.

Prediction vs. Measurement



Seeing the Minimum

The method of least squares is a minimization process.

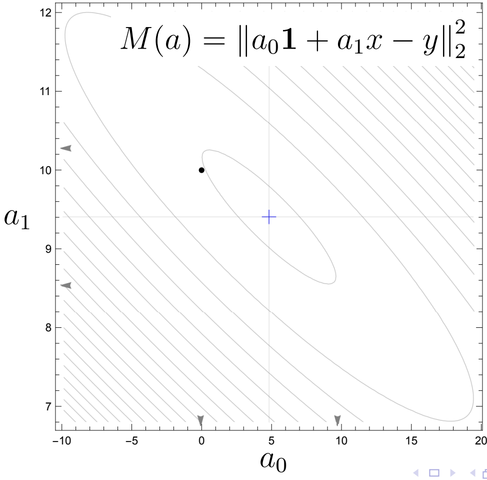
The minimization happened in a vector space of dimension 9, and cannot be viewed directly.

Seeing the Minimum

But the design matrix $\mathbf{A}$ (yet to be introduced) is a map connecting the measurement space of dimension 9 to the solution space of dimension 2.

We can see a topological map of the minimization surface and we can plot the solution (blue cross) and visually compare it to the lowest point. The black dot represents the ideal solution.

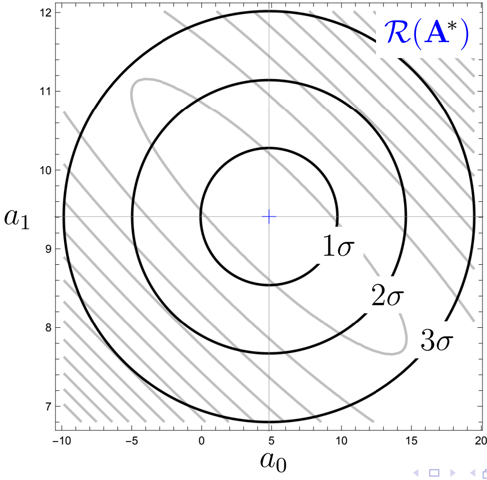
Seeing the Minimum



Seeing the Minimum

Even better, we can add error ellipsoids corresponding to confidence bands of 1σ, 2σ, and 3σ to the level curves.

Seeing the Minimum

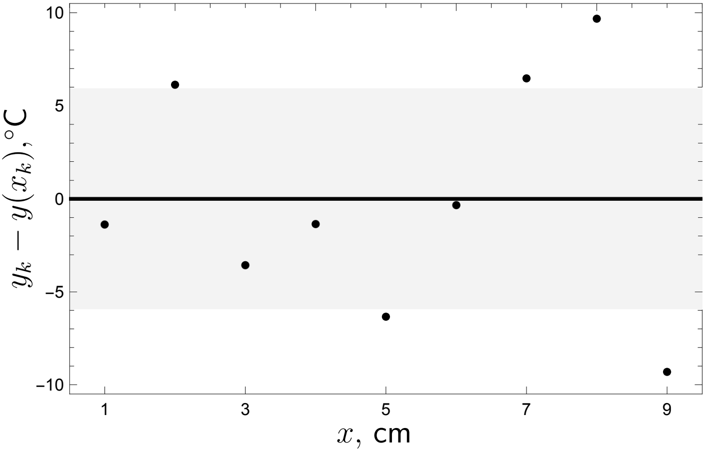


Residual Errors

We expect 68% of the errors to be contained in the 1σ error band.

That would be 6 of the 9 data points. How did we do?

Residual Errors

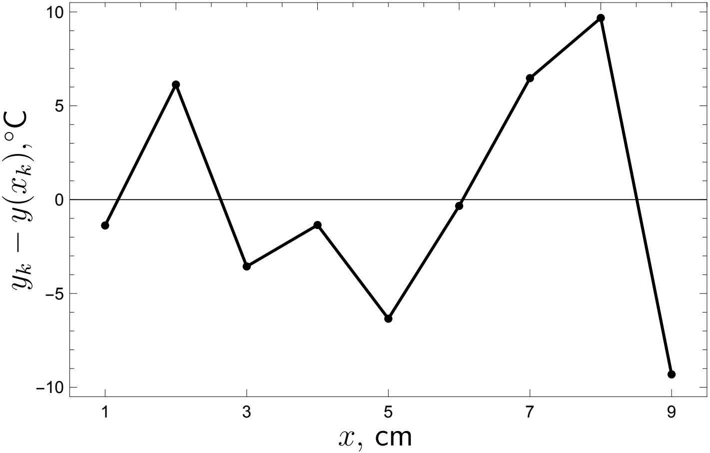


Residual Errors

The residual errors should not be correlated.

Correlated errors imply a systematic effect; the model is incomplete.

Residual Errors: Correlation

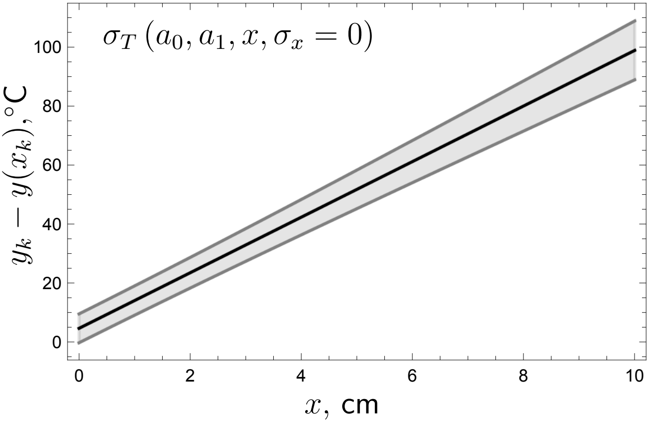


Interpolation Errors

The interpolation error combines the error in intercept and the error in slope.

The gray band represents the 1σ confidence band.

Seeing the Uncertainty

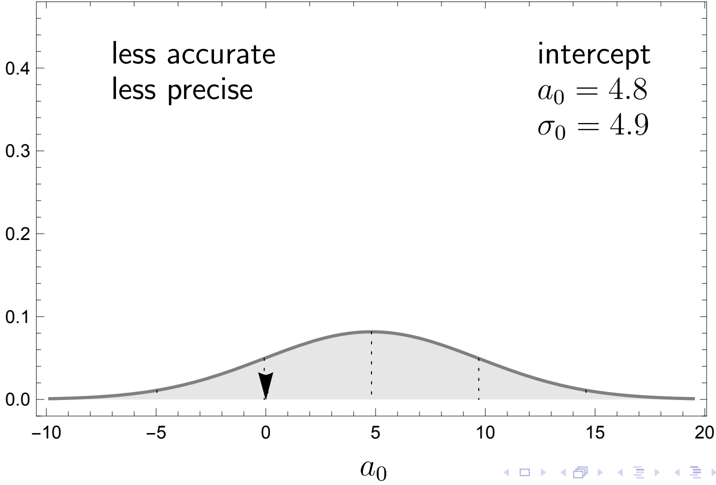


Error Distribution

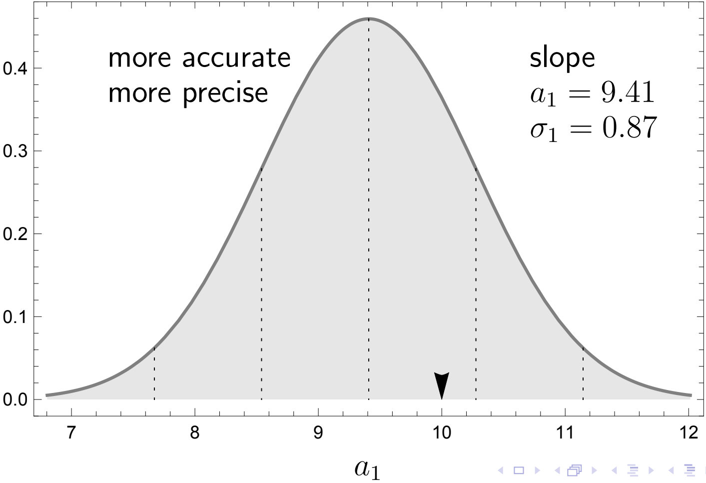
We can now view the error distributions and make quantitative comparisons.

We can make qualitative statements about precision and accuracy.

Error Distribution: Intercept



Error Distribution: Slope



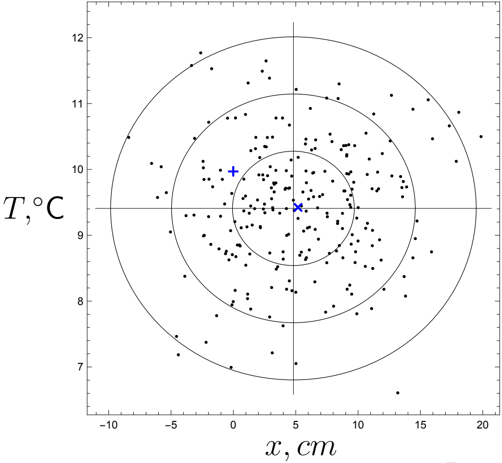
Whisker Diagram

What does it look like when we vary two parameters?.

What kind of lines are produced when we vary slope and intercept according to the normal distribution?

Let's create a sample of 250 solutions.

Whisker Diagram

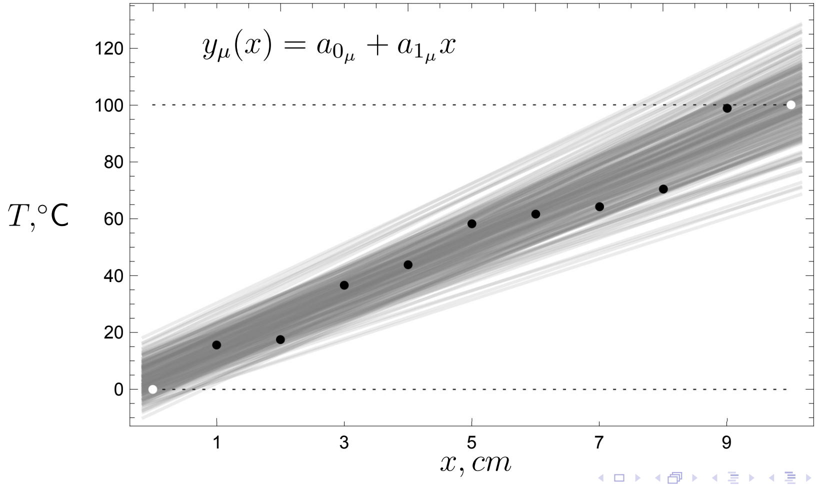


Whisker Diagram

Let's plot the curves corresponding to the 250 solutions

White dots represent the ideal solution.

Whisker Diagram

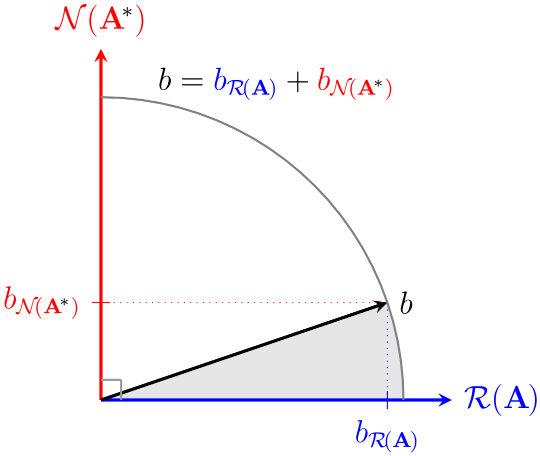


Signal To Noise

The method of least squares has extracted the signal and removed the noise.

The method of least squares has found the particular solution and removed the homogenous solution.

Signal To Noise: Geometry



Signal To Noise: Value

$$\frac{\left\| y_{\mathcal{R}(\mathbf{A})} \right\|_2^2}{\left\| y_{\mathcal{N}(\mathbf{A}^*)} \right\|_2^2} = \frac{\|y\|_2}{\|r\|_2} = \frac{13.1069}{4.21841} \approx 9.654$$

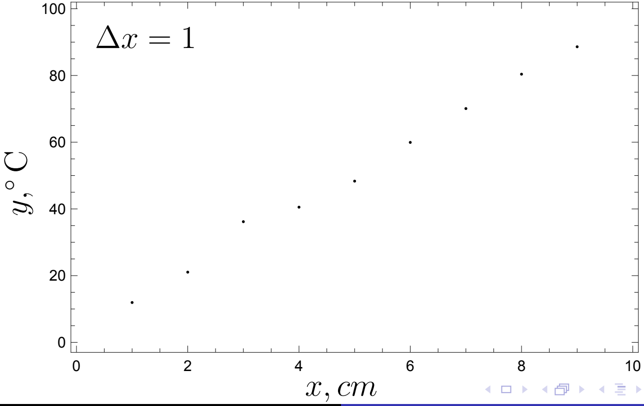
More Data = Better Precision

As we have characterized the error distribution in the slope and intercept, we have characterized the experimental apparatus.

Therefore, we can quantify the improvements in accuracy and precision we expect from larger sets of measurements.

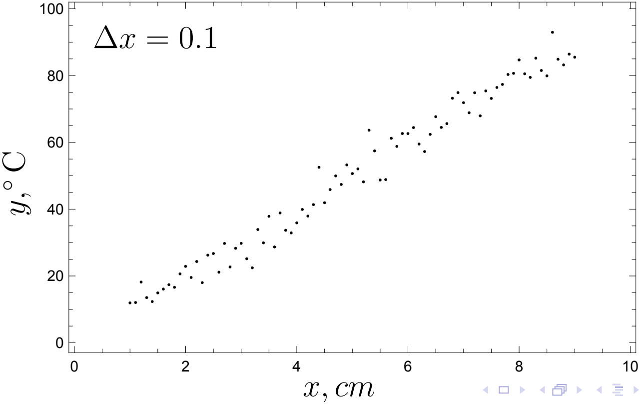
More Data = Better Precision

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 3.1 \\ 9.53 \end{bmatrix} \pm \begin{bmatrix} 1.5 \\ 0.27 \end{bmatrix}$$



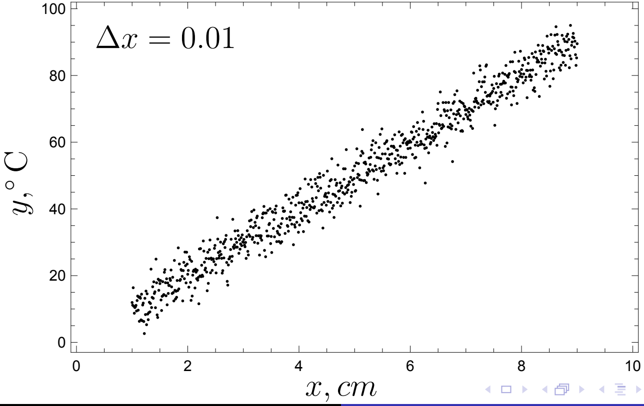
More Data = Better Precision

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} -0.4 \\ 10.03 \end{bmatrix} \pm \begin{bmatrix} 1.0 \\ 0.18 \end{bmatrix}$$



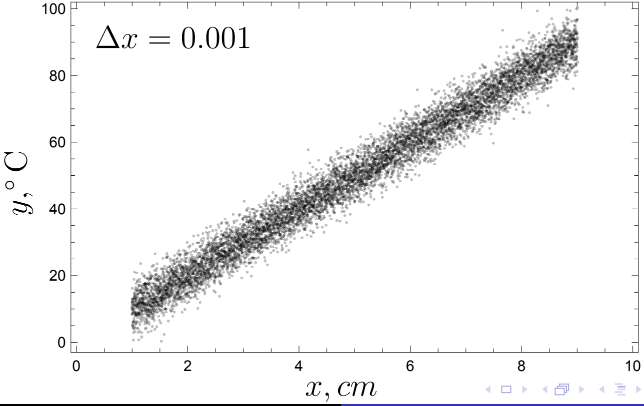
More Data = Better Precision

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} -0.12 \pm 0.34 \\ 10.011 \pm 0.061 \end{bmatrix}$$



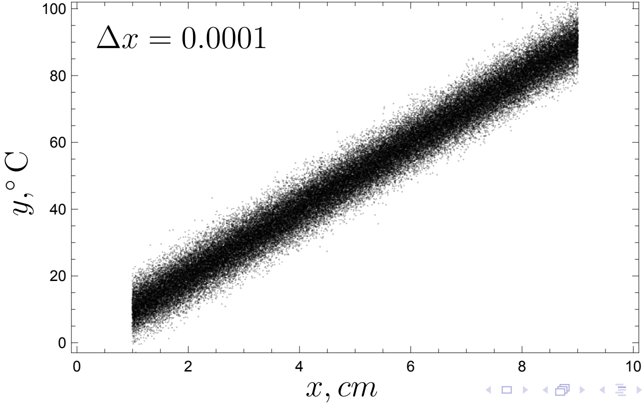
More Data = Better Precision

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 0.01 \pm 0.11 \\ 9.990 \pm 0.019 \end{bmatrix}$$



More Data = Better Precision

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} -0.029 \pm 0.034 \\ 10.0040 \pm 0.0061 \end{bmatrix}$$



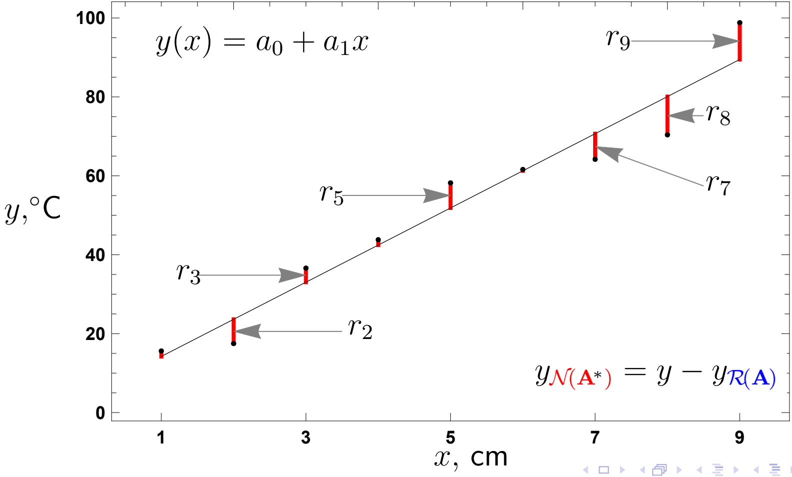
Seeing $9-$vectors

The measurement space of dimension $n = 9$ is difficult to visualize.

However, we have been seeing the 9 components of the residual error vector all along!

These are the residual errors on the solution vs. data plot, here highlighted in red.

Seeing $9-$vectors

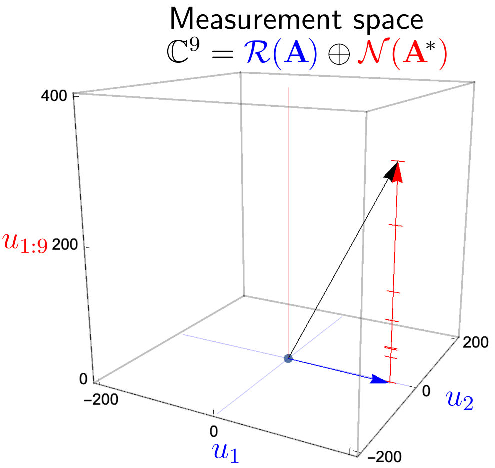


Seeing $9-$vectors

The following plot shows a decomposition of the measurement (black arrow) into a 9 dimensional space. The particular solution is drawn in a plane.

The red line represents a hyperplane of dimension 9 and is partitioned into the individual residual error components.

Seeing $9-$vectors



Vector Operations

The mathematical solution has an elegant expression in terms of vector operations.

Vector Operations

$$\begin{aligned}
\langle \mathbf{1}, \mathbf{1} \rangle &= \mathbf{1}^{\mathrm{T}}\mathbf{1} = \mathbf{1} \cdot \mathbf{1} = \sum_{k=1}^{m} 1 = \underbrace{1 + 1 + \cdots + 1}_{m \text{ instances}} = m, \\
\langle \mathbf{1}, x \rangle &= \mathbf{1}^{\mathrm{T}} x = \mathbf{1} \cdot x = \sum_{k=1}^{m} x_k, \\
\langle x, y \rangle &= x^* y = x \cdot y = \sum_{k=1}^{m} x_k \bar{y}_k.
\end{aligned} \tag{28}$$

Numeric Details

The six fundamental quantities are based upon the mesh and the data.

Numeric Details

$$
\begin{aligned}
\langle \mathbf{1}, \mathbf{1} \rangle &= 9, \\
\langle \mathbf{1}, x \rangle &= 45, \\
\langle \mathbf{1}, y \rangle &= 466.7, \\
\langle x, x \rangle &= 285, \\
\langle x, y \rangle &= 2898, \\
\Delta &= 540.
\end{aligned} \tag{29}
$$

Prediction vs. Measurement

A table of numbers is an awful way to present results.

Prediction vs. Measurement

	Measurement		Prediction	
	(cm)	(°C)	(°C)	(°C)
k	x_k	T_k	$T(x_k)$	r_k
1	1	15.6	14.2	–1.37778
2	2	17.5	23.6	6.13056
3	3	36.6	33.0	–3.56111
4	4	43.8	42.4	–1.35278
5	5	58.2	51.9	–6.34444
6	6	61.6	61.3	–0.336111
7	7	64.2	70.7	6.47222
8	8	70.4	80.1	9.68056
9	9	98.8	89.5	–9.31111
Σ	45	466.7		

Double Precision Results

The uncertainty parameters signal the significant digits. Here, an extra digit is included as a guard digit.

Double Precision Results

$$
\begin{aligned}
\sigma_0 &= 4.886206312183354 && \text{rounded to 4.9,} \\
a_0 &= 4.813888888888889 && \text{rounded to 4.8;} \\
\sigma_1 &= 0.8683016476563611 && \text{rounded to 0.87,} \\
a_1 &= 9.408333333333333 && \text{rounded to 9.41.}
\end{aligned}
\tag{30}
$$

Least Squares Result

boo

Venn Diagram of Matrix Pseudoinverses

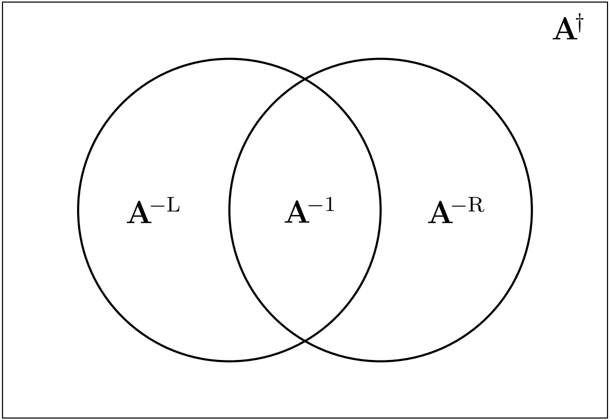


Figure: Venn diagram of the matrix pseudoinverse and specific forms.

Moore–Penrose Pseudoinverse Decompositions

type			$\mathbf{U}$	Σ	$\mathbf{V}^*$	$\mathbf{A}^\dagger$
①	$\mathbf{A}$	$=$	$\left[\begin{array}{c} \mathbf{U}_{\mathcal{R}(\mathbf{A})} \end{array}\right]$	$\left[\begin{array}{c} \mathbf{S} \end{array}\right]$	$\left[\begin{array}{c} \mathbf{V}^*_{\mathcal{R}(\mathbf{A}^*)} \end{array}\right]$	$\mathbf{A}^{-1}$
②	$\mathbf{A}$	$=$	$\left[\begin{array}{cc} \mathbf{U}_{\mathcal{R}(\mathbf{A})} & \mathbf{U}_{\mathcal{N}(\mathbf{A}^*)} \end{array}\right]$	$\left[\begin{array}{c} \mathbf{S} \\ \mathbf{0} \end{array}\right]$	$\left[\begin{array}{c} \mathbf{V}^*_{\mathcal{R}(\mathbf{A}^*)} \end{array}\right]$	$\mathbf{A}^{-\mathrm{L}}$
③	$\mathbf{A}$	$=$	$\left[\begin{array}{c} \mathbf{U}_{\mathcal{R}(\mathbf{A})} \end{array}\right]$	$\left[\begin{array}{cc} \mathbf{S} & \mathbf{0} \end{array}\right]$	$\left[\begin{array}{c} \mathbf{V}^*_{\mathcal{R}(\mathbf{A}^*)} \\ \mathbf{V}^*_{\mathcal{N}(\mathbf{A})} \end{array}\right]$	$\mathbf{A}^{-\mathrm{R}}$
④	$\mathbf{A}$	$=$	$\left[\begin{array}{cc} \mathbf{U}_{\mathcal{R}(\mathbf{A})} & \mathbf{U}_{\mathcal{N}(\mathbf{A}^*)} \end{array}\right]$	$\left[\begin{array}{cc} \mathbf{S} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{array}\right]$	$\left[\begin{array}{c} \mathbf{V}^*_{\mathcal{R}(\mathbf{A}^*)} \\ \mathbf{V}^*_{\mathcal{N}(\mathbf{A})} \end{array}\right]$	$\mathbf{A}^\dagger$

(31)

Least Squares Result

Table: Equivalent forms for the pseudoinverse matrix.

type	shape	row, col, rank	equivalence	form
①	square	$m = n = \rho$	$\mathbf{A}^{\dagger} = \mathbf{A}^{-1}$	
②	tall	$m > n,\ n = \rho$	$\mathbf{A}^{\dagger} = \mathbf{A}^{-\mathrm{L}}$	$= \left(\mathbf{A}^{*}\mathbf{A}\right)^{-1}\mathbf{A}^{*}$
③	wide	$m = \rho,\ n > m$	$\mathbf{A}^{\dagger} = \mathbf{A}^{-\mathrm{R}}$	$= \mathbf{A}^{*}\left(\mathbf{A}\mathbf{A}^{*}\right)^{-1}$
④	arbitrary	$m \ne n,$ $m \ne \rho,\ n \ne \rho$	$-$	

Pseudoinverse for a Vector

$$
\begin{aligned}
w^{\dagger} &= \quad \mathbf{V} \qquad\qquad \Sigma^{(\dagger)} \qquad\qquad \mathbf{U}^{*} \\
&= \begin{bmatrix} \mathbf{V}_{\mathcal{R}(\mathbf{A}^*)} \end{bmatrix} \begin{bmatrix} \mathbf{S}^{-1} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{U}^*_{\mathcal{R}(\mathbf{A})} \\ \mathbf{U}^*_{\mathcal{N}(\mathbf{A}^*)} \end{bmatrix} \\
&= \begin{bmatrix} 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{w^* w}} & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{w^* w}} \begin{bmatrix} \overline{w}_1 \\ \vdots \\ \overline{w}_m \end{bmatrix} \\ \mathbf{U}^*_{\mathcal{N}(\mathbf{A}^*)} \end{bmatrix} \\
&= \frac{w^*}{\|w\|_2^2},
\end{aligned} \tag{32}
$$

Pseudoinverse for a Rank 1 Matrix

Given a rank 1 matrix, $\mathbf{A} \in \mathbb{C}_1^{m \times n}$, the pseudoinverse matrix is given by

$$\mathbf{A}^{\dagger} = \frac{\mathbf{A}^{*}}{\|\mathbf{A}\|_F^2} \tag{33}$$

Fundamental Theorem of Linear Algebra

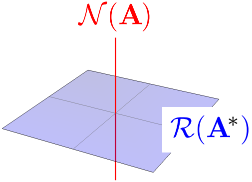
The Fundamental Theorem is an elegant statement which provides the stage upon which the drama of existence and uniqueness will unfold.

Hyperlinked references:

- Wolfram Mathworld
- Gilbert Strang
- The Four Fundamental Subspaces: 4 Lines
- Elementary Proofs

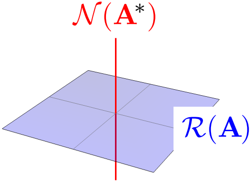
Fundamental Theorem of Linear Algebra

Row space	Mapping	Column space
Solution space		Measurement space
$n-$vectors		$m-$vectors



$$\mathbf{A} \colon \mathbb{C}^n \mapsto \mathcal{R}(\mathbf{A})$$

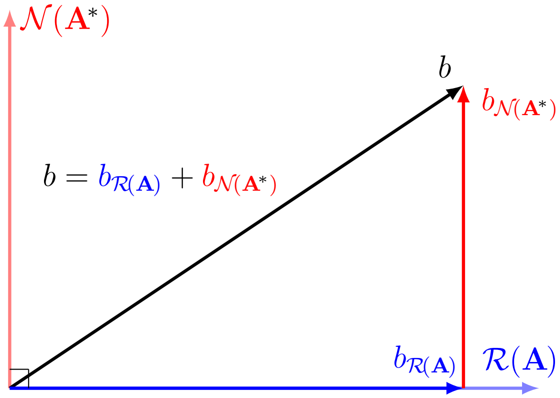
$$\mathcal{R}(\mathbf{A}^*) \leftarrow\!\!\dashv \mathbb{C}^m : \mathbf{A}^*$$



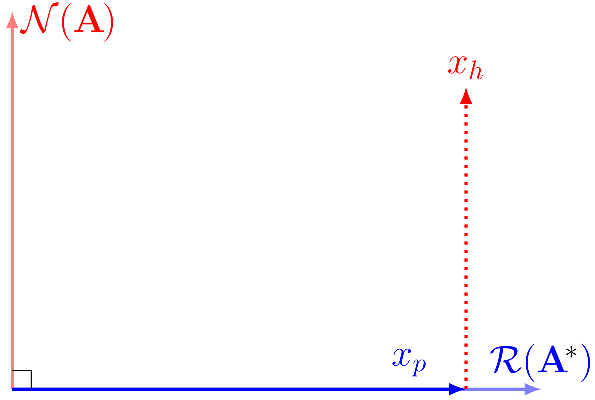
$$\mathbb{C}^n = \mathcal{R}(\mathbf{A}^*) \oplus \mathcal{N}(\mathbf{A})$$

$$\mathbb{C}^m = \mathcal{R}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^*)$$

Geometry of Least Squares Data Vector



Geometry of Least Squares Solution Vector



Least Squares Result

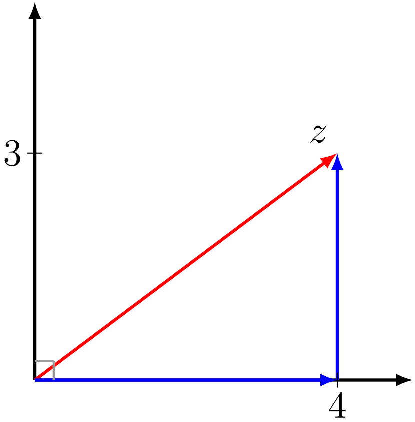
$$
\begin{aligned}
t^2 &= \left\| \mathbf{A}\, x_{LS} - b \right\|_2^2 && \text{minimization target} \\
&= \left\| \mathbf{A}(x_p + x_h) - \left(b_{\mathcal{R}(\mathbf{A})} + b_{\mathcal{N}(\mathbf{A}^*)}\right) \right\|_2^2 && \text{subspace decomposition} \\
&= \left\| \left(\mathbf{A}\, x_p - b_{\mathcal{R}(\mathbf{A})}\right) - b_{\mathcal{N}(\mathbf{A}^*)} \right\|_2^2 && \mathbf{A}\, x_h = \mathbf{0}, \text{transitivity} \\
&= \left\| -b_{\mathcal{N}(\mathbf{A}^*)} \right\|_2^2 && \mathbf{A}\, x_p = b_{\mathcal{R}(\mathbf{A})} \\
&= {b_{\mathcal{N}(\mathbf{A}^*)}}^* b_{\mathcal{N}(\mathbf{A}^*)} && \text{definition of } 2\text{−norm} \\
&= \left\langle b_{\mathcal{N}(\mathbf{A}^*)}, b_{\mathcal{N}(\mathbf{A}^*)} \right\rangle && \text{definition of } 2\text{−norm}
\end{aligned}
\tag{34}
$$

Different Notions of Length: $p-$norms

How far does a crow fly?
(Euclidean norm, Pythagorean norm, $2-$norm)

How far does a taxi drive?
(Manhattan norm, taxicab norm, $1-$norm)

Norms: Measuring Distance

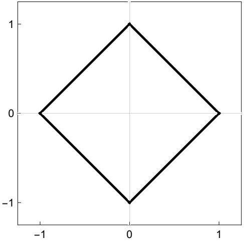
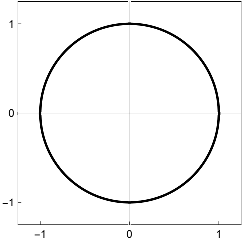
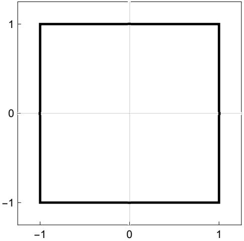


Different Notions of Length: $p-$norms

Common $p-$norms

$$
\begin{aligned}
\|p\|_1 &= |x| + |y| &= 7 \\
\|p\|_2 &= \sqrt{x^2 + y^2} &= 5 \\
\|p\|_\infty &= \max\left(|x|, |y|\right) &= 4
\end{aligned}
\tag{35}
$$

Different Notions of Length: $p-$norms

$p = 1$	$p = 2$	$p = \infty$
		
$\|x+y\|_1 = 1$	$\|x+y\|_2 = 1$	$\|x+y\|_\infty = 1$

Norm Definitions

For $p \geq 1$ the $p-$norm for $x \in \mathbb{C}^m$

$$\|x\|_p = \left(\sum_{k=1}^{m} |x_k|^p \right)^{1/p} \tag{36}$$

Norm Definitions

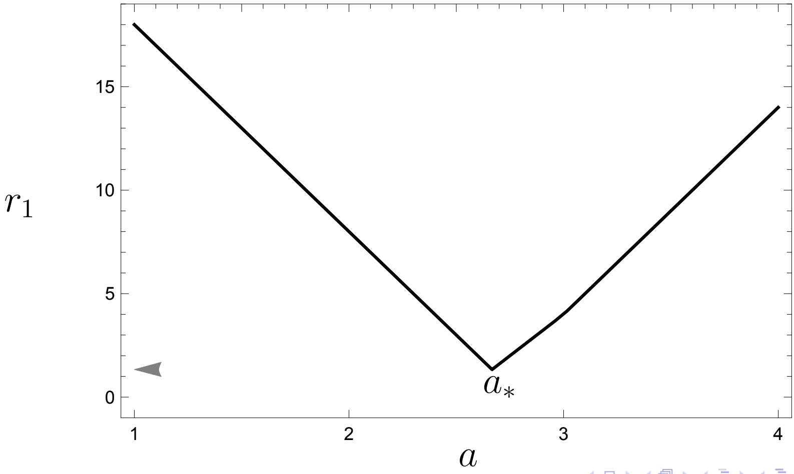
Properties of vector norms.

1. $\|x\|_p \geq 0$ with $\|x\|_p = 0 \iff x = \mathbf{0}$, (positive definite)
2. $\|\alpha x\|_p = |\alpha| \, \|x\|_p \,, \alpha \in \mathbb{C}$, (absolutely homogeneous)
3. $\|x + y\|_p \leq \|x\|_p + \|y\|_p$. (triangle inequality)

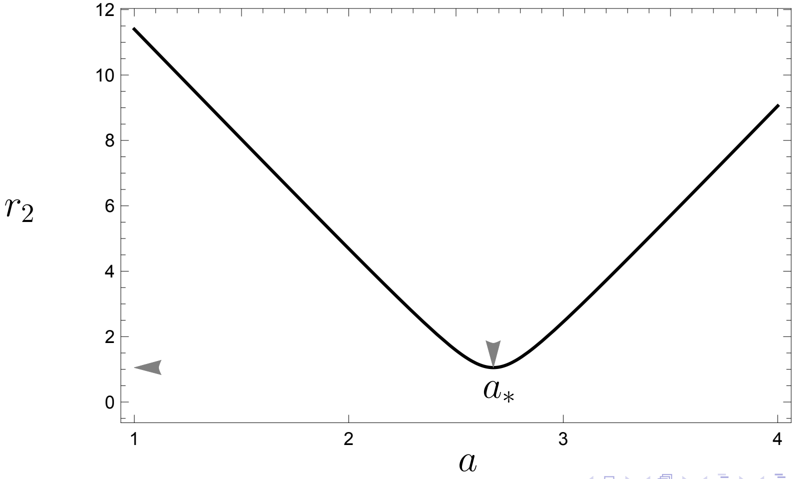
Why choose the $2-$norm

1. Gauss-Markov theorem
2. Differentiable
3. Precendence
4. Comfort

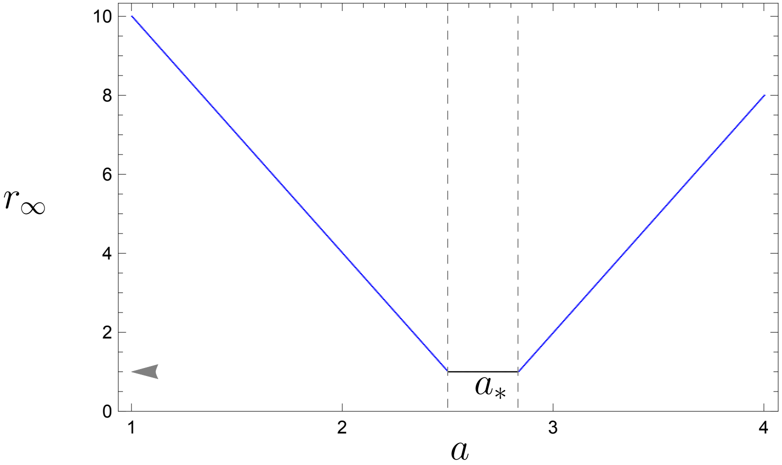
Differentiability: $1-$norm



Differentiability: $2-$norm



Differentiability: $\infty-$norm



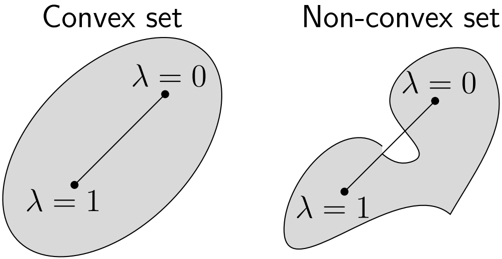
Convexity

Given a matrix $\mathbf{A} \in \mathbb{C}_{\rho}^{m\times n}$ with rank $(\rho \leq n)$, and a data vector $b \in \mathbb{C}^{m}$, then χ, the set of least squares minimizers,

$$\chi = \arg\min_{x\in\mathbb{C}^{n}} \left\| \mathbf{A}\, x - b \right\|_{2}, \tag{37}$$

is a convex set.

Seeing Convexity



Convexity

Consider two minimizers $x_1, x_2 \in \chi$ and a variation parameter $\lambda \in [0, 1]$. Establish that the linear combination is also in the set of minimizers:

$$\lambda x_1 + (1 - \lambda) x_2 \in \chi \tag{38}$$

Solve using matrix norm properties and the theorem of Pythagoras:

$$\begin{aligned} &\|\mathbf{A}\left(\lambda x_1 + (1-\lambda) x_2\right) - b\|_2^2 \leq \\ &\lambda \|\mathbf{A}\, x_1 - b\|_2^2 + (1-\lambda) \|\mathbf{A}\, x_2 - b\|_2^2 \end{aligned} \tag{39}$$

Convexity

Because the x variables are minimizers, the squared norms achieve the minimum value (least total squared error) t^2.

$$\lambda \left\| \mathbf{A} x_1 - b \right\|_2^2 + (1 - \lambda) \left\| \mathbf{A} x_2 - b \right\|_2^2 = \lambda t^2 + (1 - \lambda) t^2 = t^2 \tag{40}$$

Therefore, for all possible values of $\lambda \in [0, 1]$, the norm of the linear combination of the minimizers in (39) also maintains minimal value:

$$\left\| \mathbf{A} \left(\lambda x_1 + (1 - \lambda) x_2 - b \right) \right\|_2^2 = t^2. \tag{41}$$

Linear System

The design matrix $\mathbf{A} \in \mathbb{C}^{m\times n}_{\rho}$, the data vector $b \in \mathbb{C}^{m}$, and the solution vector $x \in \mathbb{C}^{n}$. If the equality holds, the solution is exact. Otherwise, we are finding an approximate solution.

$$\boxed{\mathbf{A}x = b} \tag{42}$$

Least Squares Problem

The definition of the least squares problem is a touchstone: x_{LS} is defined as the set of complex $n-$vectors such that the difference between prediction and measurement achieves minimum value in the $2-$norm

$$\boxed{x_{LS} = \underset{x \in \mathbb{C}^n}{\arg\min} \left\| \mathbf{A}\, x - b \right\|_2} \tag{43}$$

Decomposition of Least Squares Solution Set

Resolve the set of least squares minimizers x_{LS} into range space and null space components

$$\boxed{x_{LS} \quad = \quad x_{\mathcal{R}(\mathbf{A}^*)} \quad + \quad x_{\mathcal{N}(\mathbf{A})}} \tag{44}$$

Complete Least Squares Solution

The complete least squares solution is the typical combination of a vector $x_p \in \mathbb{C}^n$, a particular least squares solution (range space component), and a continuum solution, $x_h \in \mathbb{C}^n$, a homogeneous least squares solution (null space component).

$$\boxed{x_{LS} = x_p + x_h} \tag{45}$$

Least Squares Data Vector

Resolve the data vector b into a range space component $b_{\mathcal{R}(\mathbf{A})}$ and a null space component $b_{\mathcal{N}(\mathbf{A}^*)}$:

$$\boxed{b = b_{\mathcal{R}(\mathbf{A})} + b_{\mathcal{N}(\mathbf{A}^*)}} \tag{46}$$

Particular Least Squares Solution

Every least squares produces an exact solution and this exact solution is the particular least squares solution, x_p

$$\boxed{\mathbf{A}\, x_p = b_{\mathcal{R}(\mathbf{A})}} \tag{47}$$

Homogeneous Least Squares Solution

A linear system will present a non-trivial homogeneous least squares solution iff the right–hand null space is non-trivial. So if

$$\mathcal{N}(\mathbf{A}) \neq \{0\}, \tag{48}$$

there will be an infinite number of solutions signaled by the arbitrary vector y,

$$\boxed{x_h = (\mathbf{I}_n - \mathbf{A}^{\dagger}\mathbf{A})y = \mathbf{P}_{\mathcal{N}(\mathbf{A})}y, \qquad y \in \mathbb{C}^n} \tag{49}$$

General Least Squares Solution

Usage is a matter of preference and context. Below the line, are reminders of the fundamental subspace decomposition for the solution.

$$\boxed{\begin{array}{cccccc} x_{LS} & = & \mathbf{A}^{\dagger}b & + & (\mathbf{I}_n - \mathbf{A}^{\dagger}\mathbf{A})y \\ & = & \mathbf{A}^{\dagger}b & + & \mathbf{P}_{\mathcal{N}(\mathbf{A})}y \\ \hline & = & x_p & + & x_h \\ & = & x_{\mathcal{R}(\mathbf{A}^*)} & + & x_{\mathcal{N}(\mathbf{A})} \end{array}} \tag{50}$$

Least Squares Residual Error Vector

The residual error vector resides in the left–hand null space $\mathcal{N}(\mathbf{A}^*)$. For an arbitrary vector $x \in \mathbb{C}^n$, the residual error vector is

$$\boxed{r(x) = \mathbf{A}\,x - b} \tag{51}$$

Least Squares Merit Function

The signature property of the method of least squares the minimization of the sum of the squares of the residual errors $r^*(x)r(x)$. The merit function is the target function for minimization:

$$\boxed{M(x) = r^*(x)r(x) = (\mathbf{A}x - b)^* (\mathbf{A}x - b)} \tag{52}$$

Least Total Squared Error

The method of least squares is the collection of tools and methods which minimizes the sum of the squares of the residual errors. When this minimal value is achieved, the magnitude of the merit function represents the least total square error, t^2. This value is achieved at x_{LS}:

$$\boxed{t^2 = r^*(x_{LS})r(x_{LS})} \tag{53}$$

Least Squares Uncertainty

For the overdetermined case when $m > n$, and with full column rank $(\rho = n)$, the vector quantifying least squares uncertainty is

$$\boxed{\sigma_k^2 = \frac{t^2}{m-n} \left(\mathbf{A}^* \mathbf{A}\right)^{-1}_{k,k}, \quad k = 1 : n} \tag{54}$$

Be alert to context as the uncertainties are often called the errors in the fit parameters, or errors.

Least Squares Result

There are two styles to represent the the error terms: either as a separate vector or as part of the solution vectors:

$$\boxed{x_p = \begin{bmatrix} x_1 \\ \vdots \\ x_\rho \end{bmatrix} \pm \begin{bmatrix} \sigma_1 \\ \vdots \\ \sigma_\rho \end{bmatrix} \quad \text{or} \quad x_p = \begin{bmatrix} x_1 \pm \sigma_1 \\ \vdots \\ x_\rho \pm \sigma_\rho \end{bmatrix}} \tag{55}$$